Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/15/2024
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Effects of Low-Dose Radiation on Neurovascular Remodeling and Blood-Retina Barrier Function in Mice, Rats and Rabbits		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	12/14/2017	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per NSSC information (Ed., 8/6/24) NOTE: End date changed to 09/30/2024 per NSSC information (Ed., 11/9/23) NOTE: End date changed to 09/30/2023 per C. Ribeiro/JSC and NSSC information (Ed., 4/7/22) NOTE: End date changed to 09/30/2022 per NSSC information (Ed., 8/13/21) NOTE: End date changed to 09/30/2021 per NSSC information (Ed., 5/20/21) NOTE: End date changed to 02/28/2021 per NSSC information (Ed., 9/21/20) NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 8/8/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sun, Shu-Wei Ph.D. (Loma Linda University)		

Grant/Contract No.:	80NSSC18K0310
Performance Goal No.:	
Performance Goal Text:	
Task Description:	A recent report shows that more than 50% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. To date, the mechanisms behind these effects are not fully understood. These visual disturbances have been hypothesized to be related to increases in intraocular (IOP) or intracranial pressure (ICP). Modeling study showed that compromise in the integrity of the vascular blood-brain barrier (BBB) would serve to elevate ICP. Damage to microvascular endothelial cells induced by oxidative stress is one of the major factors thought to underlie the increased permeability of the blood-retinal barrier (BRB), which maintains a similar function in the eye for regulating intraocular pressure as the BBB in the cranium. Our knowledge about low-dose radiation risks with long-term spaceflight on retinal vasculature and their potential role in altering the BRB is very limited and inconsistent. The objectives of this application are to characterize the impact of radiation doses and qualities on retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and alteration of cell-cell interactions in the function of blood-retina-barrier (BRB). Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species will be available for sharing from three NASA funded investigators. We will evaluate radiation–induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated. Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques will provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data from established model will provide unique criteria for risks of functional detriments associated with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Task Progress:	The purpose of the study is to determine the extent to which continuous partial gravity provided by centrifugation on the International Space Station (ISS) as a countermeasure mitigates the detrimental effects of microgravity on the retina. SpaceX Commercial Resupply Services-27 (CRS-27) was launched on March 14, 2023 at the NASA Kennedy Space Center (KSC) on a 35-day mission. The eyes were evaluated in 9-week old C57BL/6 male mice after SpaceX returned from the mission using mouse Habitat Cage Unit (HCU) in the Japanese Aerospace Exploration Agency (JAXA) facility in the ISS. In this study, mice were centrifuged at different gravity levels ranging from 0 to 1g achieved by varying the rotation rate of centrifuge for the duration of a 30-day flight mission. Our in-depth transcriptomic analysis allowed us to robustly characterize and compare the cellular response in the retina and optic nerve. This study aims to evaluate whether artificial gravity (AG) could mitigate deteriorate effects of microgravity on retina and optic nerve head (ONH) during spaceflight as an appropriate countermeasure. Our data also aims to define the relationship between gravitational dose and physiological response by assessing retinal (blood-retinal barrier/BRB) biomarkers and transcriptomics endpoints. We evaluated biomarkers for oxidative damage and BRB biomarkers. 4-hydroxynonenal (4-HNE), an oxidative damage marker specific for lipid peroxidation was evaluated. A significant increase in immunoreactivity of HNE was seen in the retina of the microgravity (MG) and 0.33G mice groups compared to 0.67G, 1G, and habitat ground control (GC) groups ($p<0.05$). The levels of immunoreactivity were similar between the MG and 0.33G groups. Genomic analysis showed that many key genes responsible for cell death, cell repair, inflammation, and metabolic stress were significantly altered in MG mice compared to GC animals. Additionally, there were more significant changes in regulated gene expression in the MG group relative to that in the AG groups, especially 0.67 and 1G groups. Pattern of expression of 0.67 and 1g groups are more resembled to GC groups. These data provide evidence that spaceflight induces retinal apoptosis,oxidative stress, and changes in retinal gene expression related to cellular structure, immune response, and metabolic function, and that artificial gravity provides some protection against these changes. There is a relationship between gravitational dose and physiological response in the retina.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Seta Stanbouly and Xiao W. Mao. "Effects of spaceflight and artificial gravity on the mouse retina: a progress report. " 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Articles in Peer-reviewed Journals	Waisberg, E., Ong, J., Masalkhi, M. Xiao W. Mao, Afshin Behesti, Andrew G. Lee. "Mitochondrial dysfunction in Spaceflight Associated Neuro-Ocular Syndrome (SANS): A molecular hypothesis in pathogenesis." Eye (Lond). 2024 Jun;38(8):1409-11. https://doi.org/10.1038/s41433-024-02951-3 ; PMID: 38326485 ; PMCID: PMC11126720 , Jun-2024
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